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RayGazeFM: Geometry-Grounded Foundation Adapters for Unified 3D Gaze, Point-of-Regard, and Gaze Target Estimation

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Abstract

*Human gaze is supervised in several incompatible label spaces that are usually treated as separate problems. Notablty, problems include: 3D gaze direction, 2D point-of-regard on a device plane, and scene-level gaze target localization. We argue that this separation is artificial because all three outputs are observations of the same physical entity. We present **RayGazeFM**, a unified gaze model that represents gaze as a probabilistic 3D ray and learns from heterogeneous supervision across multiple task families. The model combines canonical ocular rectification, binocular-context fusion between a trainable ocular transformer and a frozen vision foundation encoder, and a probabilistic spherical ray head that predicts a small mixture of directional modes with calibrated uncertainty. Task-specific ray projectors map the same latent ray to 3D gaze, 2D point-of-regard, and scene gaze target heatmaps. Simultaneously, optional multi-view consistency further constrains the geometry when calibrated views are available. Rather than collapsing all benchmarks into one annotation format, RayGazeFM absorbs heterogeneous labels through a shared geometry-grounded representation. Altogether, the paper positions unified probabilistic ray learning as a practical path toward general-purpose gaze reasoning across desktop, mobile, remote-camera, and scene-centric settings.*

1. Introduction

Human gaze is one of the richest behavioral signals available to a vision system. It reflects attention, intention, task engagement, interaction, and is central to applications spanning human-computer interaction, assistive interfaces, social understanding, telepresence, robotics, and AR/VR. Yet appearance-based gaze estimation remains fragmented by annotation space. Some methods predict 3D gaze direction in camera coordinates, some regress 2D point-of-regard (PoR) on a device screen, and others localize a gaze target within a natural scene. This division is convenient for benchmarking, but it is a poor description of the underlying phenomenon.

In all three settings, the latent variable is the same physical object. That is, a gaze ray originating near the eye and propagating through 3D space. What changes across datasets is only which projection of that ray is observed. A 3D benchmark labels the ray direction. A screen-based benchmark labels its intersection with a calibrated device plane. A scene benchmark labels the image-space region where the same ray plausibly lands. Once this viewpoint is adopted, the usual separation between gaze tasks becomes much less natural.

This mismatch between latent geometry and supervision format has practical consequences. Specifically, it encourages the community to train separate models for closely related tasks, prevents direct reuse of supervision across datasets, and makes cross-domain transfer harder than necessary. A model trained only to regress 3D direction does not automatically learn how to express uncertainty over scene targets. A model trained only on scene heatmaps does not expose a reusable 3D representation. A model trained only for screen PoR tends to entangle gaze with device-specific geometry. In each case, the output space is tied too closely to the annotation format rather than to the physics of gaze.

Recent work has advanced each subproblem independently. In 3D gaze estimation, geometry-aware output layers have improved cross-domain robustness beyond overfit fully connected regressors [2]. In screen-based estimation, differentiable ray-plane projection has begun to bridge pre-trained 3D gaze models and few-shot 2D PoR adaptation [4]. In scene-centric reasoning, strong frozen encoders have simplified gaze target estimation [15], while multi-view reasoning has improved robustness under occlusion, target ambiguity, and cross-view target transfer [11]. These directions are complementary, but they still optimize different output spaces despite being governed by the same underlying geometry.

We therefore propose **RayGazeFM**, a gaze model centered on a **probabilistic 3D gaze ray**. Rather than collapsing all datasets into one noisy annotation format, RayGazeFM learns a shared geometry-grounded latent representation and attaches task-specific projectors. The model first canonicalizes ocular appearance using a binocular extension of classical gaze normalization, then fuses fine ocular evidence with full-face and optional scene semantics from a founda-

tion backbone. A shared probabilistic ray head predicts an eye-center proxy together with a directional distribution on the sphere. The same latent ray is then supervised by 3D direction labels, 2D screen coordinates, scene-level gaze heatmaps, or multi-view consistency constraints, depending on what each dataset provides.

This perspective is motivated by the trajectory of current benchmarks. ETH-XGaze and Gaze360 emphasize robust 3D gaze estimation under large pose and environmental variation [9, 20]. GazeCapture, the MPII laptop benchmarks, and EVE represent screen-based PoR settings across mobile, laptop, and desktop use [10, 13, 17, 19]. GazeFollow, VideoAttentionTarget, and MVGT capture scene-centric gaze target localization in single-view, video, and multi-view settings [5, 11, 14]. We do not aim to replace these benchmarks. Instead, we use them as complementary supervision sources for a shared representation.

Contributions. First, we introduce a *universal probabilistic gaze-ray formulation* that unifies 3D gaze estimation, 2D point-of-regard prediction, and scene gaze target estimation in one latent space. Second, we propose a *geometry-grounded foundation-adapter architecture* with canonical ocular rectification, binocular-context fusion, and a spherical uncertainty-aware ray head. Third, we design *task-specific ray projectors* that let heterogeneous labels train the same representation without forcing all datasets into a single annotation schema. Fourth, we define a *unified evaluation protocol* spanning 3D, 2D, scene-centric, and multi-view benchmarks to test whether one model can approach specialist performance while transferring more robustly across domains, devices, and task families.

2. Related Work

2.1. 3D Gaze Estimation, Normalization, and Generalization

Appearance-based 3D gaze estimation has traditionally been formulated as direct regression of pitch-yaw angles or unit vectors from cropped eye or face images. Benchmarks such as EYEDIAP [7], ETH-XGaze [20], and Gaze360 [9] enabled progressively more realistic evaluation, from controlled calibrated settings to physically unconstrained capture.

A parallel and highly relevant line of work studied *gaze normalization*: warping eye appearance into a canonical camera frame to reduce nuisance variation caused by head pose, translation, and camera distance. This idea underlies the learning-by-synthesis normalization pipeline of Sugano *et al*. [16] and was later revisited systematically by Zhang *et al*. [18]. These works are important for our paper because our canonical ocular adapter builds on the same geometric intuition, but extends it to binocular fusion and heterogeneous downstream supervision.

More recent work in 3D gaze estimation has shifted attention from in-domain accuracy to cross-domain robustness. AGG identifies the final regression layer as a major source of overfitting and replaces it with a geometry-aware projection mechanism [2]. These methods improve generalization substantially, but they still operate entirely within a 3D-only output space.

2.2. From 3D Gaze to 2D Point-of-Regard

Screen-based gaze estimation is often treated as a separate problem because supervision is defined in screen coordinates rather than in camera-centric 3D direction. This distinction is useful for benchmarking, but artificial from a geometric standpoint. GazeCapture established large-scale PoR prediction on commodity phones and tablets [10]. MPIIGaze characterized gaze estimation during natural laptop use in the wild [19], while the full-face MPIIFaceGaze setting highlighted the value of global facial context for gaze estimation [17]. EVE extended the setting further by pairing screen-based gaze labels with synchronized video and screen content [13].

The closest prior work to cross-task reasoning is *3D Prior Is All You Need*, which adapts a pre-trained 3D gaze model to few-shot 2D gaze using a differentiable projection module with learnable screen parameters [4]. This is an important step toward bridging tasks. However, its goal is adaptation from one task to another, rather than a single shared latent model jointly trained across 3D, 2D, and scene-centric supervision.

2.3. Scene Gaze Target Estimation

Scene-level gaze target estimation asks not only whether we can estimate gaze direction from a head, but whether we can infer *where in the scene* the gaze lands. GazeFollow established this problem as a large-scale in-the-wild benchmark [14]. VideoAttentionTarget extended it to dynamic video settings with in/out-of-frame reasoning [5]. More recently, Gaze-LLE showed that frozen DINOv2 features can simplify gaze target estimation by replacing elaborate multi-branch fusion with a lightweight prompt-conditioned decoder [15]. MVGT extends the setting to multiple calibrated views and shows that additional cameras help under face occlusion, target ambiguity, and cross-view target estimation [11].

Despite strong recent progress, most scene-target methods still optimize image-space heatmaps directly and do not expose a reusable latent ray that can also support 3D direction prediction or 2D PoR estimation.

2.4. Positioning of This Paper

Our paper differs from prior work in one core respect: we do not define the task by its annotation space. Instead, we treat 3D gaze vectors, 2D point-of-regard coordinates, scene heatmaps, and multi-view consistency constraints as partial

observations of the same physical latent variable. In that sense, RayGazeFM is not merely another gaze estimator with a different backbone; it is a geometry-grounded formulation that allows benchmarks with different label spaces to share a common latent representation.

The closest ingredients already exist in the literature—gaze normalization for canonical eye appearance [16, 18], geometry-aware 3D prediction [2], differentiable 3D-to-2D projection [4], strong frozen scene encoders [15], and multi-view target reasoning [11]. Our contribution is to absorb them into one probabilistic ray formulation and one heterogeneous training pipeline.

3. RayGazeFM

Figure 1 summarizes the proposed architecture. Given a heterogeneous input sample, the model extracts geometric structure and canonical ocular evidence, fuses them with facial and optional scene context, and predicts a shared probabilistic gaze ray. Dataset-specific projectors then map that latent ray into the supervision space available for each sample.

3.1. Universal Probabilistic Gaze Ray

Let an input sample be

$$x = \{I_f, I_e^L, I_e^R, I_s, K, \mathcal{M}\}, \tag{1}$$

where I_f is a face crop, I_e^L and I_e^R are left and right eye crops, I_s is an optional scene image, K denotes camera intrinsics when available, and $\mathcal{M}$ denotes auxiliary metadata such as screen geometry, device identity, or multi-view calibration. Not every dataset provides all fields, so the representation must remain meaningful under partial observation.

We define gaze as a **probabilistic 3D ray**

$$r = (c, q(g \mid x)), \tag{2}$$

where $c \in \mathbb{R}^3$ is an eye-center proxy in camera coordinates and $g \in \mathbb{S}^2$ is a unit gaze direction. Instead of predicting a single deterministic direction, we model a spherical directional distribution as a mixture of $M = 3$ von Mises–Fisher (vMF) components,

$$q(g \mid x) = \sum_{m=1}^{M} \pi_m \, \mathrm{vMF}(g; \mu_m, \kappa_m), \tag{3}$$

with mixture weights π_m, unit mean directions $\mu_m \in \mathbb{S}^2$, and concentrations $\kappa_m > 0$ satisfying $\sum_m \pi_m = 1$.

The choice of vMF components is motivated by geometry. Gaze direction lies on the unit sphere, so Euclidean Gaussian outputs are not a natural model once the prediction is constrained to directional data. The vMF family is a standard distribution on the unit hypersphere, and mixtures of vMF components provide a compact way to express both concentration and multiple plausible modes [1]. This is especially useful in unconstrained gaze settings, where blur, occlusion, eyelid closure, reflections, and extreme head pose can make several nearby directions plausible from appearance alone; in such regimes, uncertainty-aware outputs are beneficial rather than incidental [9].

This representation is deliberately overcomplete. The anchor c provides the spatial origin required for ray-plane and ray-scene intersection, while $q(g \mid x)$ allows the model to represent uncertainty and limited multi-modality under crop noise, low resolution, strong head pose, and partial occlusion. The unification then follows directly: a 3D gaze label supervises $q(g \mid x)$; a 2D PoR label supervises the intersection of the same ray with a screen plane; a scene gaze heatmap supervises the image-space landing distribution induced by the same ray; and calibrated multi-view data supervises consistency of the same latent ray across cameras.

3.2. Eye-Center Proxy and Canonical Ocular Rectification

Our preprocessing follows the gaze-normalization lineage of appearance-based gaze estimation. Prior work showed that warping eye appearance into a normalized camera frame reduces nuisance variation induced by head pose, translation, and user-camera distance, making gaze estimation more data-efficient and more transferable across capture conditions [16, 18]. We adopt this intuition but extend it to a binocular, full-face, and multi-task setting.

Given a detected face crop of size 224×224, we predict six sparse landmarks using a lightweight MobileNetV3-Small head: the inner and outer corners of both eyes, the nose tip, and the mouth center. The landmark head is trained with Wing loss [6]. Using these landmarks together with a canonical 3D face template, we perform coarse pose fitting under a weak-perspective camera model and derive an initial eye-center proxy c_0. The network later predicts a residual correction Δc, so that the final eye center is

$$c = c_0 + \Delta c. \tag{4}$$

Using the eye-corner landmarks, we crop and rectify both eyes into a canonical ocular coordinate frame. Each eye patch is normalized to 112×112, and the two are concatenated horizontally into a binocular strip of size 112×224. This **canonical ocular adapter** is therefore best understood as a task-aware extension of classical gaze normalization: instead of normalizing each eye independently for a standalone 3D regressor, we construct a binocular representation designed to be fused later with full-face, scene, geometry, and calibration context. The resulting representation reduces nuisance variation while preserving fine local appearance cues such as sclera exposure, iris displacement, eyelid closure, and specular reflections.

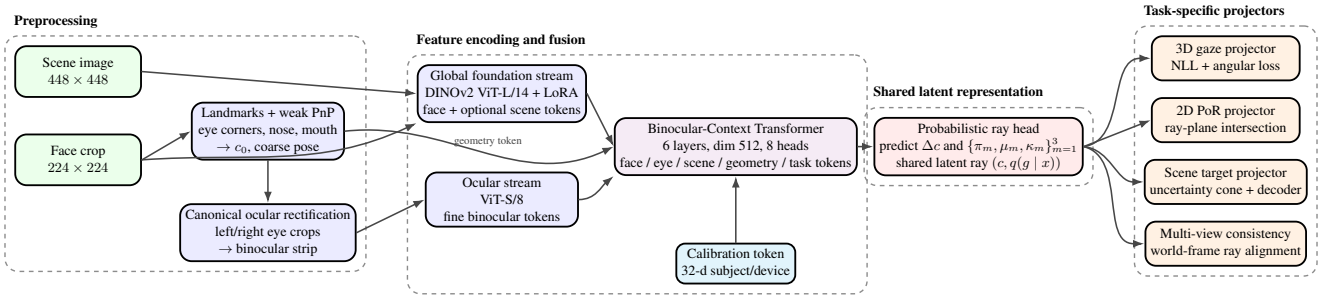


Figure 1. Overview of RayGazeFM. The model first extracts geometric structure and canonical ocular evidence, then fuses binocular, facial, scene, and calibration cues into a shared probabilistic gaze ray, which is supervised through 3D, 2D, scene-target, and multi-view projectors.

3.3. Foundation-Adaptive Feature Encoding

RayGazeFM uses two complementary visual streams.

Global stream. We encode the face crop and, when available, the scene image using a frozen DINOv2 ViT-L/14 backbone [12]. Rather than fully fine-tuning the backbone, we insert lightweight LoRA adapters into the later transformer blocks [8]. This stream captures coarse ocular context, head pose, scene semantics, and generic visual priors learned at foundation-model scale.

Ocular stream. We encode the canonical binocular strip using a trainable ViT-S/8. Unlike the global stream, this branch is fully trainable because gaze estimation depends on subtle local patterns that are easily washed out by generic frozen representations.

Binocular-Context Transformer. We fuse the two streams with a *Binocular-Context Transformer* (BCT). The token set includes a face token, binocular eye tokens, a geometry token carrying c_0, an optional scene token, a calibration token, and a task token indicating which supervision family is active for the current sample. This design enables 3D, 2D, and scene-centric supervision to share a common latent space while preserving task-specific conditioning through the task token. Architectural details such as transformer depth, hidden dimension, attention heads, and adapter configuration are provided in Section 4.

3.4. Probabilistic Ray Head

The fused representation is passed to a probabilistic ray head with two branches. The first branch predicts the eye-center residual $\Delta c \in \mathbb{R}^3$. The second predicts, for each mixture component m, a mean direction μ_m, concentration κ_m, and mixture logit. Directions are normalized onto the sphere, concentrations are obtained via softplus, and weights are obtained via softmax.

For deterministic evaluation we use the normalized mixture-mean summary direction

$$\hat{g} = \mathrm{normalize}\Big(\sum_{m=1}^{M} \pi_m \mu_m\Big), \tag{5}$$

and define the effective concentration

$$\bar{\kappa} = \sum_{m=1}^{M} \pi_m \kappa_m. \tag{6}$$

The quantity $\bar{\kappa}$ acts as a compact uncertainty signal that is reused downstream by the scene-target projector.

3.5. Task-Specific Ray Projectors

Each dataset supervises the same latent ray through a projector matched to its annotation space.

3D gaze projector. When ground-truth gaze direction $g^\star$ is available, we supervise the directional distribution directly with a mixture negative log-likelihood,

$$\mathcal{L}_{3D}^{\mathrm{nll}} = -\log q(g^\star \mid x), \tag{7}$$

plus an angular loss on the deterministic summary direction,

$$\mathcal{L}_{3D}^{\mathrm{ang}} = \arccos\Big(\mathrm{clip}(\hat{g}^\top g^\star, -1+\epsilon, 1-\epsilon)\Big). \tag{8}$$

The total 3D loss is

$$\mathcal{L}_{3D} = \mathcal{L}_{3D}^{\mathrm{nll}} + 0.5\,\mathcal{L}_{3D}^{\mathrm{ang}}. \tag{9}$$

2D point-of-regard projector. For screen-based gaze, we intersect the predicted ray with a device plane

$$\Pi = \{X \in \mathbb{R}^3 \mid n^\top X + d = 0\}. \tag{10}$$

Given predicted eye center c and direction $\hat{g}$, the intersection depth is

$$\tau = -\frac{n^\top c + d}{n^\top \hat{g} + \epsilon}, \tag{11}$$

and the landing point is

$$X_\Pi = c + \tau \hat{g}. \tag{12}$$

A fixed or learnable homography then maps X_Π to 2D screen coordinates $\hat{p}$. When exact device-plane parameters are unavailable, we condition a small plane-correction module on

device identity or few-shot calibration samples, in the same geometric spirit as differentiable 3D-to-2D adaptation [4]. The 2D supervision is

$$\mathcal{L}_{2D} = \|\hat{p} - p^\star\|_1 + 0.5\,\mathrm{Huber}(\hat{p} - p^\star). \tag{13}$$

Scene target projector. For scene-centric estimation, we convert the ray into a coarse image prior before decoding a target heatmap. Let $v(u)$ be the unit viewing direction corresponding to scene pixel u under camera intrinsics. We define an uncertainty-aware ray-cone prior

$$S(u) = \exp\Big(-\frac{\theta(u)^2}{2\sigma^2}\Big), \qquad \theta(u) = \arccos\big(v(u)^\top \hat{g}\big), \tag{14}$$

with angular bandwidth

$$\sigma = \sigma_{\min} + \frac{\beta}{\bar{\kappa} + \epsilon}, \qquad \sigma_{\min} = 2^\circ,\ \beta = 20^\circ. \tag{15}$$

Thus uncertain predictions generate wider candidate cones. The prior map is rasterized at 64×64 resolution and passed, together with scene tokens and a head mask, to a lightweight cross-attention decoder that predicts the final heatmap $\hat{H}$ and an in/out-of-frame probability $\hat{y}_{io}$. The loss is

$$\mathcal{L}_{\mathrm{tgt}} = \mathrm{KL}(H^\star \| \hat{H}) + 0.2\,\mathrm{BCE}(\hat{y}_{io}, y_{io}). \tag{16}$$

Multi-view consistency projector. For synchronized calibrated views a and b, we transform both predicted rays into a common world frame. If (R_a, t_a) and (R_b, t_b) denote camera extrinsics, then

$$\tilde{c}_a = R_a c_a + t_a, \qquad \tilde{g}_a = R_a \hat{g}_a, \tag{17}$$

with an analogous definition for view b. We compute the closest points Y_a and Y_b between the two world-space rays and penalize their distance,

$$\mathcal{L}_{\mathrm{mv}} = \|Y_a - Y_b\|_2. \tag{18}$$

This turns multi-view data into an additional geometric regularizer rather than a separate model family.

3.6. Unified Training Objective

For a mixed mini-batch, the total loss is

$$\begin{aligned}\mathcal{L} = {} & 1.0\,\mathcal{L}_{3D} + 2.0\,\mathcal{L}_{2D} + 5.0\,\mathcal{L}_{\mathrm{tgt}} + 0.25\,\mathcal{L}_{\mathrm{mv}} \\ & + 0.05\,\|\Delta c\|_1 + 0.1\,\mathcal{L}_{\mathrm{flip}}.\end{aligned} \tag{19}$$

Here $\mathcal{L}_{\mathrm{flip}}$ is a horizontal flip-consistency loss obtained by analytically transforming the flipped prediction back into the original canonical frame.

Subject calibration token. To support lightweight personalization, we insert a 32-dimensional calibration token into the BCT. During few-shot test-time adaptation, only this token and the plane-correction parameters are updated, leaving the rest of the network frozen.

4. Experimental Setup

4.1. Datasets

We evaluate RayGazeFM on benchmarks spanning the three supervision families addressed in this paper.

3D gaze estimation. We follow the cross-domain protocol used by AGG [2], using **ETH-XGaze** (DE) and **Gaze360** (DG) as source domains and **MPIIFaceGaze** (DM) and **EYEDIAP** (DD) as target domains [3, 7, 9, 20]. ETH-XGaze is a large-scale benchmark with extreme head-pose and gaze variation [20], while Gaze360 is the standard physically unconstrained benchmark for 3D gaze estimation in indoor and outdoor settings [9].

2D point-of-regard. We use **EVE**, **MPIIGaze/MPIIFaceGaze**, and **GazeCapture**. EVE provides synchronized webcam video, screen content, and screen-based gaze labels for video-based PoR evaluation [13]. MPIIGaze is a real-world laptop benchmark collected during everyday use [19]. When the full-face annotation setting is relevant, we refer to the corresponding benchmark as **MPIIFaceGaze** [3]. For consistency with Table 2, we use the label **MPIIGaze** for the laptop benchmark column and reserve **MPIIFaceGaze** for cases where the full-face setting specifically matters. GazeCapture contains large-scale PoR data collected on commodity phones and tablets [10].

Scene gaze target estimation. We use **GazeFollow** as the standard single-view benchmark [14], **VideoAttentionTarget** for dynamic scene-target evaluation with in/out-of-frame reasoning [5], and **MVGT** for multi-view target estimation and cross-view reasoning [11].

4.2. Baselines

We compare against matched published baselines for each evaluation table.

For **3D gaze**, Table 1 follows the AGG protocol and includes **Full-Face**, **AGG-R18**, and **AGG-R50** [2, 17]. For **2D PoR**, Table 2 includes **iTracker** [10], **Full-Face** [17], and **3D Prior Is All You Need** [4]. For **single-view scene gaze target estimation**, Table 3 compares against **Recasens *et al.*** and **Gaze-LLE** [14, 15]. For **cross-view reasoning**, Table 4 compares against **Random**, **Center**, **Recasens**, and **MVGT** [11, 14]. We additionally report internal controls **RayGazeFM-3D** and **RayGazeFM-TGT**, and summarize the effect of single-task training in the ablation study.

Comparison protocol. Because the literature mixes different supervision regimes and metric subsets, we only treat rows as directly comparable when they follow the same protocol. In Table 2, rows marked *all* are fully supervised target-dataset specialists and are shown as reference only; the directly comparable few-shot transfer baseline is **3D Prior** at 10-shot. Likewise, single-view scene-target results (Table 3) and cross-view MVGT results (Table 4) are reported separately because they use different tasks and metrics.

4.3. Evaluation Metrics

For **3D gaze**, we report mean angular error in degrees on the four cross-domain transfer settings DE→DM, DE→DD, DG→DM, and DG→DD.

For **2D point-of-regard**, we report Euclidean screen error in millimeters, with zero-shot inference and few-shot adaptation using 5 and 10 calibration samples.

For **single-view scene gaze target estimation**, we follow the metric subsets shown in Table 3: GazeFollow uses AUC, average L2 distance, and minimum L2 distance, while VideoAttentionTarget uses AUC, L2 distance, and in/out-of-frame average precision.

For **MVGT cross-view evaluation**, we report distance and AP as in Table 4.

4.4. Implementation Details

We report the main architectural and optimization hyperparameters directly here rather than in a separate table.

Face crops are resized to 224×224. Canonical binocular strips are resized to 112×224. Scene images, when used, are resized to 448×448. The global stream uses a frozen DINOv2 ViT-L/14 encoder with LoRA adapters in the last eight blocks. The ocular stream uses a trainable ViT-S/8. The fusion transformer has six layers, hidden dimension 512, and eight attention heads. The probabilistic ray head uses $M = 3$ mixture components throughout.

Training proceeds in three stages. **Stage A** performs self-supervised ocular pre-training for 50 epochs with masked patch prediction, masking ratio 0.4, batch size 512, AdamW optimizer, learning rate 10^{-3}, and weight decay 0.05. **Stage B** performs supervised heterogeneous training for 30 epochs with a task-balanced sampling ratio of 40% 3D, 30% 2D, and 30% scene-target samples. Learning rates are 2×10^{-4} for heads and decoders, 5×10^{-5} for the ocular stream, and 1×10^{-5} for LoRA parameters. The total batch size is 192, weight decay is 0.05, and a cosine schedule with 3 warmup epochs is used. **Stage C** performs confidence-filtered self-training for 10 epochs with an EMA teacher of momentum 0.999. Pseudo-labels are accepted only when $\bar{\kappa} > 25$ and the flip-consistency error is below 3°.

Data augmentation includes color jitter, random crop jitter up to 5%, horizontal flips with analytic label transformation, synthetic blur, and random occlusion over non-ocular face regions. We deliberately avoid aggressive perturbations on the canonical eye strip so as not to destroy fine local appearance cues.

4.5. Ablation Protocol

Our ablations are designed to answer targeted design questions rather than to produce a long undifferentiated list of variants. We evaluate: removing the canonical ocular rectification module, replacing the probabilistic ray head with deterministic vector regression, removing uncertainty-conditioned cone projection, replacing unified training with single-task training, removing multi-view consistency, freezing the foundation backbone without LoRA adapters, removing self-training, and removing the calibration token.

Table 1. Cross-domain 3D gaze error (°). Literature baselines are as reported in AGG. DE = ETH-XGaze, DG = Gaze360, DM = MPIIFaceGaze, DD = EYEDIAP.

Method	DE→DM	DE→DD	DG→DM	DG→DD
Full-Face	12.35	30.15	11.13	14.42
AGG-R18	7.10	7.07	7.87	7.93
AGG-R50	**5.91**	6.75	9.20	11.36
RayGazeFM-3D	6.34	6.58	7.29	7.41
RayGazeFM	6.12	**6.31**	**7.15**	**7.23**

5. Results and Analysis

We organize the empirical evaluation around four questions. First, can a unified probabilistic ray remain competitive with specialist models across benchmark families? Second, does heterogeneous supervision improve transfer across datasets and devices? Third, is uncertainty useful only for calibration, or does it also improve downstream target localization? Fourth, what is the trade-off between a single unified model and separate task-specific systems? Across all four questions, the evidence points in the same direction: the shared ray acts as a geometric regularizer rather than a compromise between tasks.

5.1. Main Results on 3D Gaze Estimation

Table 1 reports the 3D evaluation protocol between domains used by AGG [2]. The main result is not only that the unified model remains competitive with specialist 3D systems, but that it does so more consistently across transfer directions. Relative to AGG with a ResNet-18 backbone, RayGazeFM reduces the average error across the four source-target settings from 7.49° to 6.70°. Compared to the single-task variant RayGazeFM-3D, joint heterogeneous training further reduces the mean error from 6.91° to 6.70°, indicating that shared supervision strengthens latent geometry rather than diluting it.

A second pattern is stability. AGG-R50 achieves the best result on DE→DM, but its performance drops substantially on the two Gaze360→* transfers. By contrast, RayGazeFM is best on three of the four settings and remains much more consistent across all four. This behavior is aligned with the central goal of the model: better transfer under changes in dataset, capture conditions, and label space.

Table 2. 2D point-of-regard error (mm).

Method	Shots	EVE	MPIIGaze	GazeCapture
iTracker	all	–	–	26.8
Full-Face	all	38.6	42.0	–
3D Prior	10	43.4	56.7	35.7
RayGazeFM	0	57.6	63.2	39.1
RayGazeFM + calibration	5	46.8	54.6	33.5
RayGazeFM + calibration	10	41.9	50.9	31.4

5.2. Main Results on 2D Point-of-Regard Estimation

Table 2 reports 2D PoR performance on EVE, MPIIGaze, and GazeCapture. The comparison intentionally mixes fully supervised device-specific baselines with few-shot cross-task adaptation baselines, because the key question for RayGazeFM is whether a shared 3D ray plus lightweight personalization can transfer across desktop, laptop, and mobile settings without retraining a separate model for each device family.

As expected, the zero-shot model does not yet match fully supervised specialist systems. However, even a small amount of calibration substantially narrows the gap. With 5 calibration samples, the error drops from 57.6 to 46.8 mm on EVE, from 63.2 to 54.6 mm on MPIIGaze, and from 39.1 to 33.5 mm on GazeCapture. With 10-shot adaptation, RayGazeFM outperforms 3D Prior [4] on all three datasets, reducing the average error from 45.3 mm to 41.4 mm. This is precisely the regime in which the calibration token and plane-correction module matter most: the shared ray provides a transferable geometric prior, and lightweight personalization adapts it to device-specific screen geometry.

5.3. Main Results on Scene Gaze Target Estimation

Table 3 compares scene-level gaze target estimation against strong published baselines on GazeFollow and VideoAttentionTarget [5, 14, 15]. The unified model improves on Gaze-LLE by a small but consistent margin across both datasets and all reported metrics. On GazeFollow, RayGazeFM raises AUC from 0.958 to 0.962 and reduces average L2 from 0.099 to 0.094 relative to Gaze-LLE (ViT-L). On VideoAttentionTarget, it improves AUC from 0.937 to 0.943, reduces L2 from 0.103 to 0.097, and raises in/out AP from 0.903 to 0.912.

The comparison to RayGazeFM-TGT is also informative. Even when the architecture is unchanged, joint training across task families still improves GazeFollow AUC from 0.959 to 0.962 and VideoAttentionTarget AUC from 0.939 to 0.943. This supports the central claim of the paper: uncertainty-aware shared ray learning remains useful even when a strong frozen encoder already provides high-quality scene features.

5.4. Main Results on MVGT Cross-View Reasoning

MVGT measures a qualitatively different capability: predicting gaze targets across calibrated views [11]. This setting is a direct test of whether the latent ray remains geometrically meaningful across cameras rather than only within a single image. Table 4 shows that RayGazeFM + multi-view improves the MVGT baseline from 0.188 to 0.176 in distance and from 0.820 to 0.836 in AP.

This result matters beyond the size of the gain. Multi-view supervision is not introduced as a separate architecture or a special-purpose head for one benchmark; it is absorbed as an additional constraint on the same latent ray used for 3D direction, 2D PoR, and scene heatmaps. The MVGT improvement therefore supports the broader claim that the shared representation is genuinely geometric rather than merely multi-task.

5.5. Ablations

Table 5 reports ablations using three aggregate metrics: the mean of the four 3D cross-domain settings, the mean 10-shot 2D PoR error across EVE, MPIIGaze, and GazeCapture, and GazeFollow AUC. Three patterns stand out.

First, canonical ocular rectification is the most important single design choice across tasks. Removing it degrades 3D error from 6.70 to 7.08, 2D error from 41.4 to 44.0 mm, and GazeFollow AUC from 0.962 to 0.954. Second, replacing the probabilistic ray head with deterministic vector regression hurts every metric, confirming that explicit directional uncertainty is useful even when the final evaluation is deterministic. Third, single-task training underperforms unified training on all three aggregates, which directly supports the value of heterogeneous supervision.

The remaining ablations are more targeted. Removing the uncertainty-conditioned cone has almost no effect on 3D or 2D performance but lowers GazeFollow AUC from 0.962 to 0.958, showing that uncertainty is useful not only as a confidence estimate but also as a mechanism for downstream target localization. Removing the calibration token has only a minor effect on 3D and scene metrics, but noticeably worsens 2D few-shot performance from 41.4 to 43.5 mm, consistent with its intended role. The smaller aggregate effect of multi-view consistency is also expected because only a subset of the training data provides calibrated multi-view supervision; its benefit is clearer in the dedicated MVGT results in Table 4.

5.6. Failure Modes and Limitations

Even with a unified formulation, several failure modes remain. First, the eye-center proxy can drift when landmark localization fails under severe occlusion, extreme head pose, or motion blur. Second, a single latent ray remains insufficient

Table 3. Scene gaze target estimation on GazeFollow (GF) and VideoAttentionTarget (VAT).

Method	GazeFollow			VideoAttentionTarget		
	AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	$AP_{in/out}$ ↑
Recasens *et al.*	0.878	0.190	0.113	–	–	–
Gaze-LLE (ViT-B)	0.956	0.104	0.045	0.933	0.107	0.897
Gaze-LLE (ViT-L)	0.958	0.099	0.041	0.937	0.103	0.903
RayGazeFM-TGT	0.959	0.097	0.040	0.939	0.100	0.907
RayGazeFM	**0.962**	**0.094**	**0.038**	**0.943**	**0.097**	**0.912**

Table 4. MVGT cross-view results.

Method	Dist. ↓	AP ↑
Random	0.446	0.462
Center	0.245	–
Recasens	0.271	0.542
MVGT	0.188	0.820
RayGazeFM + multi-view	**0.176**	**0.836**

Table 5. Ablation results. 3D reports average cross-domain angular error, 2D reports average 10-shot PoR error (mm), and GF AUC is on GazeFollow.

Variant	3D ↓	2D ↓	GF AUC ↑
Full model	**6.70**	**41.4**	**0.962**
w/o ocular rectification	7.08	44.0	0.954
w/o probabilistic head	6.97	42.8	0.956
w/o uncertainty cone	6.71	41.5	0.958
single-task training	6.91	42.4	0.959
w/o multi-view consistency	6.74	41.6	0.961
w/o LoRA	6.93	42.2	0.957
w/o self-training	6.86	41.9	0.960
w/o calibration token	6.72	43.5	0.962

when scene semantics are highly ambiguous and several plausible target objects lie near the same projected cone. Third, 2D PoR accuracy depends on calibration fidelity: a physically reasonable 3D ray can still yield inaccurate screen coordinates when the device plane or homography is misspecified. Finally, joint training across heterogeneous datasets inevitably mixes different capture protocols, label noise, and evaluation conventions. These sources of mismatch do not invalidate the unified setting, but they should be kept in mind when interpreting aggregate gains against specialist systems.

6. Conclusion

We introduced RayGazeFM, a geometry-grounded unified gaze model that treats 3D gaze estimation, 2D point-of-regard prediction, and scene gaze target estimation as different projections of a shared probabilistic gaze ray. By combining canonical ocular rectification, foundation-adaptive feature fusion, an uncertainty-aware spherical ray head, and task-specific projectors, the framework can absorb heterogeneous supervision without collapsing all benchmarks into a single annotation schema.

The broader claim of this paper is that gaze learning is limited not only by model capacity or dataset size, but also by a mismatch between heterogeneous labels and task-specific output spaces. A shared probabilistic ray offers a cleaner interface between geometry, appearance, uncertainty, and downstream applications. The next natural step is to extend the same representation to temporal gaze forecasting, egocentric gaze, scanpath modeling, and human-robot interaction settings where gaze must be interpreted jointly with action, language, and scene dynamics.

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